

Devayush Rout

Applied AI Engineer | Retrieval & Operational AI Systems

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Summary

Applied AI engineer building retrieval-aware systems for enterprise workflows and operational decision support. Built high-reliability RAG systems, procurement intelligence pipelines, and multimodal compliance systems emphasizing reliability and controlled generation.

Projects

Internal Policy Intelligence System [↗](#)

- Developed a containerized RAG pipeline with owner-scoped retrieval and deterministic routing (SAFE / REFUSE / ESCALATE) for internal policy automation.
- Engineered hybrid retrieval pipelines combining semantic search, metadata filtering, and version-aware document indexing in pgvector.
- Implemented evaluation workflows across 100+ retrieval test cases, achieving Recall@5 of 0.88 and 85% verdict accuracy.

Tender Match [↗](#)

- Built async ingestion pipelines handling automated tender collection, metadata extraction, and deduplication across government portals.
- Engineered a hybrid filtering and semantic retrieval system combining SQL-based deterministic filtering with vector search for manufacturer matching.
- Reduced manual tender review cycles from ~2 hours/day to ~10 minutes through automated ranking and prioritization infrastructure.

MedWaste Guardian [↗](#)

- Deployed a multimodal compliance system via FastAPI combining YOLOv8 classification, retrieval-based regulation lookup, and voice-enabled guidance.
- Built retrieval pipelines for biomedical waste regulation grounding using RAG-based compliance lookup.
- Developed low-latency voice interaction workflows for sterile clinical environments supporting real-time disposal assistance and compliance reasoning.

Skills

AI Systems: RAG, Information Retrieval, LLM Evaluation, Document Intelligence, NLP

Backend & Infrastructure: FastAPI, PostgreSQL, pgvector, Docker

Retrieval: FAISS, ChromaDB, LlamaIndex

Languages: Python, SQL

Computer Vision: YOLOv8, OpenCV

Education

Vellore Institute of Technology, B.Tech in Computer Science and Engineering – Bhopal, MP, India

Sept 2023 – present